

GLASGOW PRESTWICK, United Kingdom

WMO: 031350

Lat: 55.509N

Lon: 4.587W

Elev: 20

StdP: 101.09

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	-2.9	-6.4	2.2	-2.3	-5.1	2.5	-1.1	17.9	6.8	16.0	7.6	0.9	110	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	23.0	17.2	21.1	16.4	19.8	15.8	18.1	21.8	17.2	20.1	16.3	18.9	3.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.0	20.0	15.9	11.3	19.0	15.0	10.7	17.9	51.1	21.9	48.2	20.3	45.8	18.8	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.5	10.1	DB	-6.3	26.7	2.1	2.4	-7.9	28.4	-9.1	29.8	-10.3	31.2	-11.8	32.9	(4)
(5)			WB	-6.9	19.8	2.2	1.2	-8.5	20.6	-9.8	21.3	-11.0	22.0	-12.6	22.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	4.9	5.1	6.2	8.4	10.9	13.6	15.3	15.1	13.2	10.4	7.2	5.0
	DBStd	4.70	3.07	2.90	2.64	2.63	2.72	2.24	2.03	2.04	2.26	2.71	3.16	3.79
	HDD10.0	776	158	139	118	61	20	1	0	0	3	28	91	156
	HDD18.3	3181	416	372	375	299	229	144	98	103	154	246	333	412
	CDD10.0	646	1	1	2	12	49	108	164	159	99	41	8	2
	CDD18.3	8	0	0	0	0	0	1	4	3	1	0	0	0
	CDH23.3	57	0	0	0	0	5	11	25	15	1	0	0	0
	CDH26.7	6	0	0	0	0	0	1	4	1	0	0	0	0
Wind	WSAvg	4.3	5.1	5.0	4.7	4.2	4.1	3.9	3.6	3.8	4.1	4.4	4.5	4.7
Precipitation	PrecAvg	949	91	65	73	51	54	61	69	85	103	110	105	96
	PrecMax	1210	183	180	135	131	122	142	164	172	245	203	182	194
	PrecMin	705	12	4	18	4	7	21	14	18	21	35	26	22
	PrecStd	125	43	39	35	25	26	23	27	43	50	50	43	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	13.0	14.9	19.2	23.2	25.0	26.8	25.8	21.2	17.8	15.0	13.2
		MCWB	10.6	11.0	10.4	12.5	15.5	17.4	18.9	18.7	16.6	14.8	13.4	11.6
	2%	DB	11.1	11.2	12.2	15.9	20.1	21.2	23.0	22.0	19.8	16.1	13.8	12.2
		MCWB	9.6	9.5	9.4	11.3	14.3	16.0	17.7	17.5	16.0	14.0	12.2	10.6
	5%	DB	10.1	10.2	11.1	14.0	17.2	19.2	21.0	20.0	18.0	15.2	12.8	11.1
		MCWB	8.9	8.7	8.8	10.5	12.9	15.1	16.8	16.5	15.2	13.2	11.2	9.7
	10%	DB	9.2	9.2	10.1	12.8	15.8	17.9	19.1	18.9	17.0	14.2	11.9	10.1
		MCWB	8.2	7.9	8.3	9.7	12.1	14.1	15.6	15.6	14.4	12.3	10.3	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	11.4	11.5	13.2	16.6	17.9	19.6	19.4	17.9	15.4	13.5	12.2
		MCDB	12.3	12.9	13.7	17.3	22.0	22.2	24.2	23.9	20.7	17.0	14.7	13.4
	2%	WB	10.1	10.1	10.3	11.9	14.9	16.6	18.3	17.9	16.5	14.6	12.4	11.1
		MCDB	10.8	11.0	11.8	15.3	19.0	20.5	22.4	21.0	18.6	16.0	13.6	12.4
	5%	WB	9.1	9.1	9.4	10.9	13.5	15.5	17.2	17.0	15.5	13.7	11.4	10.1
		MCDB	9.8	10.1	10.7	13.5	16.7	18.9	20.2	19.5	17.5	15.2	12.7	11.2
	10%	WB	8.2	8.1	8.6	10.1	12.4	14.6	16.1	16.1	14.7	12.8	10.5	9.1
		MCDB	9.2	9.2	10.0	12.2	15.2	17.6	18.6	18.2	16.6	14.3	11.7	10.1
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.6	6.3	7.3	8.0	7.2	7.0	6.9	6.6	6.0	5.5	5.4
		MCDBR	4.6	5.4	7.1	9.4	11.3	9.8	10.2	8.9	7.9	6.1	5.1	5.3
		MCWBR	4.5	4.7	5.2	6.0	6.7	5.8	5.9	5.4	5.2	4.9	4.7	5.1
	5% WB	MCDBR	4.6	4.8	6.0	8.2	10.0	9.0	8.7	7.7	6.6	5.6	5.1	5.4
		MCWBR	4.6	4.7	4.9	5.7	6.3	5.7	5.5	5.2	5.0	4.9	4.9	5.2
Clear-Sky Solar Irradiance	taub	0.338	0.348	0.373	0.390	0.393	0.388	0.389	0.373	0.365	0.348	0.328	0.323	
	taud	2.202	2.296	2.271	2.260	2.301	2.352	2.356	2.411	2.429	2.402	2.289	2.183	
	Ebn at Noon	550	713	785	828	848	857	848	840	793	709	568	469	
	Edh at Noon	61	79	101	118	120	115	113	100	87	70	55	49	
All-Sky Solar Radiation	RadAvg	0.49	1.18	2.31	3.74	4.78	4.70	4.51	3.65	2.54	1.37	0.66	0.37	
	RadStd	0.07	0.13	0.17	0.29	0.44	0.51	0.45	0.27	0.20	0.19	0.07	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	+0.71	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page